SECTION A

[48 marks]

*Answer **all** the questions in this section. All questions carry equal marks.*

1. A binary operation $*$ is defined on the set of real numbers, R, by $x*y=2x+2y-\frac{xy}{5}$, find the:
   (a) Inverse of x under the operation $*$.
   (b) Truth set when $m*7=-2*m$

2. Given that ${}^{x}C_2+{}^{x}C_3-4x=0$, where x is a positive integer, find the value of x.

3. A function f is defined by $f(x)=\frac{72}{px+r}$, where p and r are constants. Given that $f(6)=12$ and $f(7)=9$ find;
   (a) The values of p and r.
   (b) $f^{-1}\left(\frac{5}{3}\right)$

4. Resolve $\frac{2x-7}{25x-24-6x^2}$ into partial fractions.

5. The table shows the distribution of the weights (*Newtons*) of 100 blood donors.

Weight (N)	$55-59$	$60-65$	$66-71$	$72-78$	$79-86$	$87-90$
Number of donors	18	5	20	22	27	8

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   Draw a histogram for the distribution.

6. If two fair dice are thrown together twice, find the probability of obtaining a product of six in the first throw and a sum of eight in the second throw.

7. A light inextensible string passes over a smooth pulley and carries masses 4 kg and 3 kg at its ends. If the masses are released from rest, calculate:
   (a) their acceleration
   (b) their speed after 3 *seconds.*

8. A force of F*N* acts on a body of mass 50 kg and changes its velocity from 30 ms^{-1} and 40 ms^{-1} in 6 *seconds.* Find the
   (a) magnitude of the force,
   (b) acceleration of the body,
   (c) distance covered in 9 *seconds* when the initial velocity is 30 ms^{-1}.

SECTION B

[52 marks]

Answer **four** *questions* **only** *from this section with at least one from* **each** *part.*
All questions carry equal marks.

PART I
PURE MATHEMATICS

9. (a) Given that $\int_{-1}^{m}(2x^2 - x - 3)\,dx = \frac{-9}{2}$, where m is an integer, find the value of m.

   (b) Solve $2(\log_3 x - 1) = \log_3 x$ and $y = \sqrt{x} + 1$ simultaneously.

10. Given that ${}^8C_x$, ${}^7C_x$ and $\frac{7}{6}\left({}^6C_x\right)$ form the three consecutive terms of an exponential sequence $(G.P)$, find the:

   (a) Value of x;

   (b) Common ratio of the sequence;

   (c) Sum of the first ten terms of the sequence.

11. (a) Express $\left[\frac{4+\sqrt{7}}{4-\sqrt{7}}\right]^2 - \left[\frac{4-\sqrt{7}}{4+\sqrt{7}}\right]^2$ in the form $p + q\sqrt{7}$, where p and q are real numbers

   (b) Two functions f and g are on the set of real numbers by $f : x \rightarrow \sqrt{\frac{16-x^2}{x^2+4}}$ and $g : x \rightarrow \frac{x^2-4}{8-x^2}$, $x \neq \pm 2\sqrt{2}$. Find:

      (i) the domain of f;

      (ii) $g \circ f(x)$

PART II
STATISTICS AND PROBABILITY

12. If the mean and standard deviation of the numbers $x, 2.5, 4, y$ and 7 are 4 and $\sqrt{\frac{31}{10}}$ respectively. Find the value of x and y where $x < y$.

13. An association is made up of 10 journalists and 9 engineers. If a committee of 4 members is to be formed from the association, find the probability that it will consist of;

   (a) **at least two** engineers;

   (b) 75% journalists;

   (c) odd number of engineers

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PART III
VECTORS AND MECHANICS

14. (a) Five forces acting on a particle are represented by $(2i + 3j)$, $(4i - 7j)$, $(-5i + 8j)$, $(i + j)$ and $(pi + qj)$, where $p, q \in R$. If the particle is in equilibrium, find the value of p and q

(b) $WXYZ$ is a parallelogram whose vertices are $W(-1, 1)$, $X(m, n)$, $Y(5, -7)$ and $Z(2, 4)$ Find the;

(i) values of m and n;

(ii) unit vector in the direction of $\overrightarrow{XY}$.

15. An object is thrown vertically upwards from the top of a building 20m high. If the object passes the point it was thrown after 4 *seconds* on its way down, find the:

(a) velocity at which the object was thrown;

(b) time taken when the object is 10m above the level it was thrown;

(c) time taken when the object is 10m below the level it was thrown;

(d) velocity with which the object hits the ground.

[Take g = $10\, ms^{-2}$]

END OF PAPER

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